Creacio d’una imatge desde zero



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1. Crearem una imatge on hi hagi el necessari per executar l'aplicació web.
Podem buscar si trobem una imatge que ja contingui tot el necessari o bé ens la podem construir.
(Per exemple, hi ha una imatge anomenada php:apache que conté un apache amb el php integrat).
En qualsevol cas haurem d'adaptar-la a les nostres necessitats, que seran:

- **Fer servir el port 8888 per la web, tant en el contenidor com en la màquina host. Penseu**
  que heu de canviar la configuració d'Apache a la imatge. Feu-ho amb el Dockerfile.
- **Els logs d'Apache els guardarem en la màquina host. Això no cal que ho feu amb Dockerfile,**
  configureu-ho en el moment d'executar la imatge (... run -v ....)
- **La còpia dels fitxers de la pàgina haurem d'aconseguir que es faci mitjançant el Dockerfile.**

Dockerfile:

```
FROM php:8.1-apache

RUN docker-php-ext-install mysqli && docker-php-ext-enable mysqli

RUN sed -i 's/^Listen 80$/Listen 8888/' /etc/apache2/ports.conf \
 && sed -i 's/<VirtualHost \*:80>/<VirtualHost *:8888>/'
/etc/apache2/sites-available/000-default.conf

COPY index.php /var/www/html/

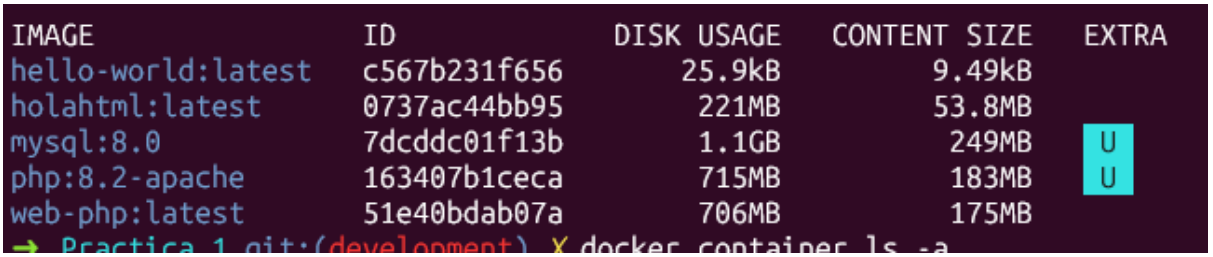
EXPOSE 888
```

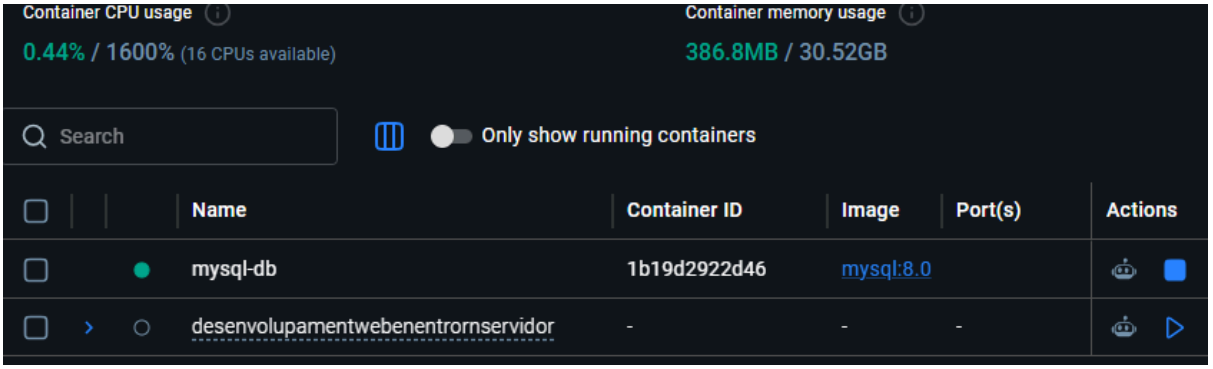
```
➜ Practica 1 git:(development) X docker build -t web-php .
[+] Building 1.2s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 340B
 => [internal] load metadata for docker.io/library/php:8.1-apache
 => [auth] library/php:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/4] FROM docker.io/library/php:8.1-apache@sha256:72f0c0134922985022d55a39b0390
 => => resolve docker.io/library/php:8.1-apache@sha256:72f0c0134922985022d55a39b0390
 => [internal] load build context
 => => transferring context: 1.36kB
 => CACHED [2/4] RUN docker-php-ext-install mysqli && docker-php-ext-enable mysqli
 => CACHED [3/4] RUN sed -i 's/^Listen 80$/Listen 8888/' /etc/apache2/ports.conf  &&
 => CACHED [4/4] COPY index.php /var/www/html/
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:17e527cb2bb844768584e57f71a25d21d20e039e05fdc154fd4
 => => exporting config sha256:c40345a292da878924f626a8e35fdd140f52392e4d3901727bda7
 => => exporting attestation manifest sha256:2549919c23be1d3c56e6eafc175ac612e022c28
 => => exporting manifest list sha256:51e40bdab07a4fe757725cd03b8d7e8b6f835210cb06eb
 => => naming to docker.io/library/web-php:latest
 => => unpacking to docker.io/library/web-php:latest
➜ Practica 1 git:(development) X
```

2. Necessitem una imatge que contingui el gestor de base de dades. Si bé hi ha opcions que les podriem configurar amb un Dockerfile, en aquest cas tot serà per la línia de comandaments.
Penseu que caldrà configurar el gestor de base de dades (contrasenyes, rutes, etc.) i crear la base de dades. Aquesta base de dades la guardarem en local (host), això vol dir que haureu de trobar en quins directoris emmagatza les dades i les configurarcions la base de dades.
Recordeu que una imatge descarregada no es pot modificar, només ho podeu fer amb un contenidor, el qual es crea amb docker run.
Mostreu una captura de docker images i docker container ls -a

```
→ Practica 1 git:(development) X docker images

IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   c567b231f656       25.9kB         9.49kB
holahtml:latest      0737ac44bb95        221MB         53.8MB
mysql:8.0            7dcddc01f13b        1.1GB          249MB   U
php:8.2-apache       163407b1ceca        715MB          183MB   U
web-php:latest       51e40bdab07a        706MB          175MB
→ Practica 1 git:(development) X docker container ls -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS                     PORTS                    NAMES
1b19d2922d46   mysql:8.0      "docker-entrypoint.s…"   4 minutes ago   Up 4 minutes               3306/tcp, 33060/tcp      mysql-db
d09c4ff1565b   php:8.2-apache "docker-php-entrypoi…"   6 weeks ago     Exited (0) 5 weeks ago                              Dev_web_servidor
c960719a13e0   mysql:8.0      "docker-entrypoint.s…"   6 weeks ago     Exited (0) 5 weeks ago                              desenvolupamentwebenent
→ Practica 1 git:(development) X
```

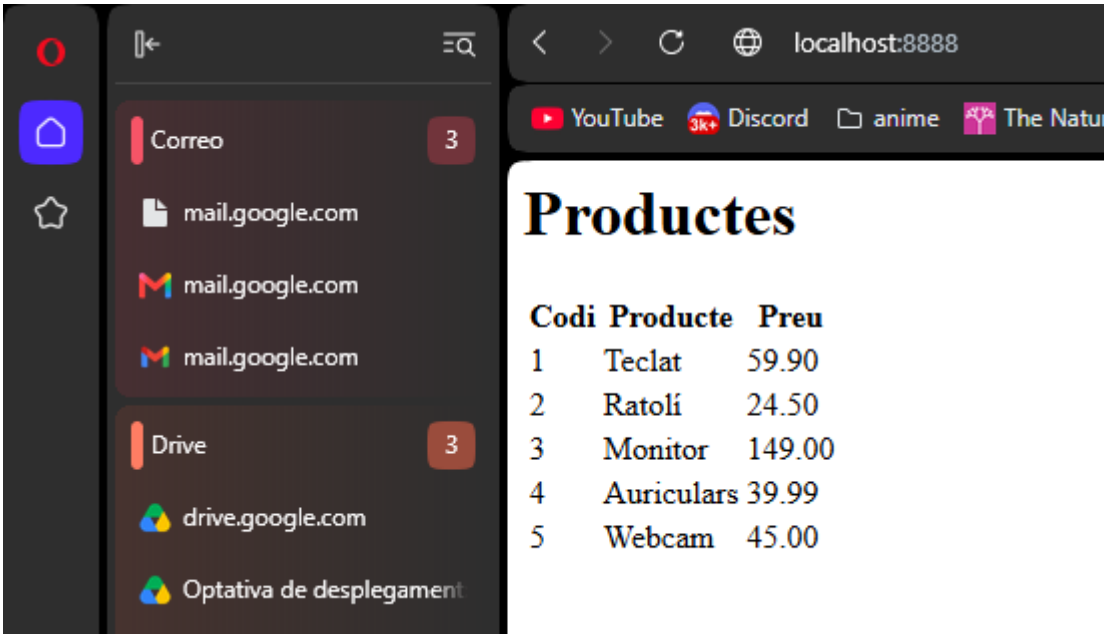




3. Engegueu les imatges. Tingueu en compte que una depèn de l'altra, per la qual cosa haureu de pensar en quin ordre les posareu en funcionament.
Comproveu que funciona: això vol dir que amb un navegador et pots connectar a la pàgina de l'aplicació i que aquesta pot accedir a la base de dades.

```
→ Practica 1 git:(development) X docker container inspect mysql-db | grep '"IPAddress"'
                    "IPAddress": "172.17.0.2",
→ Practica 1 git:(development) X
```

```
→ Practica 1 git:(development) X docker run -d --name web-php \
  -p 8888:8888 \
  -v "$PWD/logs":/var/log/apache2 \
  web-php
37ef98fe337dfd8bd15dd923f01c8d83f9b66726bf7bd272c5ddc4aeb62fd5bb
→ Practica 1 git:(development) X
```




```
→ Practica 1 git:(development) X docker exec -it mysql-db mysql --default-character-set=utf8mb4 -ppepito
mysql: [Warning] Using a password on the command line interface can be insecure.
+-----+------------+--------+
| idp | nomarticle | preu   |
+-----+------------+--------+
|   1 | Teclat     |  59.90 |
|   2 | Ratolí     |  24.50 |
|   3 | Monitor    | 149.00 |
|   4 | Auriculars |  39.99 |
|   5 | Webcam     |  45.00 |
+-----+------------+--------+
→ Practica 1 git:(development) X ls -l logs/
total 8
-rw-r--r-- 1 root root 464 Nov 10 19:16 access.log
-rw-r--r-- 1 root root 868 Nov 10 19:16 error.log
-rw-r--r-- 1 root root   0 Nov 10 19:13 other_vhosts_access.log
→ Practica 1 git:(development) X
```

4. Feu una còpia de seguretat dels contenidors que hagis fet servir i les dades internes i externes que hagi generat.

```
→ Practica 1 git:(development) X mkdir -p backup
→ Practica 1 git:(development) X docker commit web-php  web-php:backup
sha256:a6a903d8cfd76df5ff6067fa9dbf91cd8f603245b6151a684c5148766048b330
→ Practica 1 git:(development) X docker commit mysql-db mysql-db:backup
docker save -o backup/web-php-backup.tar  web-php:backup
sha256:a6fcba95cd378f168a7e94988d7b02228476b6e24b158e42fe75617db821192b
→ Practica 1 git:(development) X docker save -o backup/web-php-backup.tar  web-php:backup
^[[A^[[A^[[B^[[B^Z
[1]  + 104332 suspended  docker save -o backup/web-php-backup.tar web-php:backup
→ Practica 1 git:(development) X docker commit mysql-db mysql-db:backup
sha256:9a15dd0987e497d62a8de08cbb812fe1769fe029c3ccded5601b754bc022d5f9
→ Practica 1 git:(development) X docker save -o backup/web-php-backup.tar  web-php:backup
→ Practica 1 git:(development) X docker save -o backup/mysql-db-backup.tar mysql-db:backup
→ Practica 1 git:(development) X
```

```
→ Practica 1 git:(development) X sudo tar czf backup/mysql-data.tar.gz mysql-data
[sudo: authenticate] Password:
→ Practica 1 git:(development) X sudo tar czf backup/logs.tar.gz logs
→ Practica 1 git:(development) X ls -lh backup/
total 396M
-rw-r--r-- 1 root   root    719 Nov 10 19:19 logs.tar.gz
-rw-r--r-- 1 root   root   5.3M Nov 10 19:19 mysql-data.tar.gz
-rw------- 1 esteve esteve 223M Nov 10 19:18 mysql-db-backup.tar
-rw------- 1 esteve esteve 168M Nov 10 19:18 web-php-backup.tar
```

5. Esborra els contenidors i les imatges. Restaura-les de nou i comprova que funcioni.

Per esborrar-les:

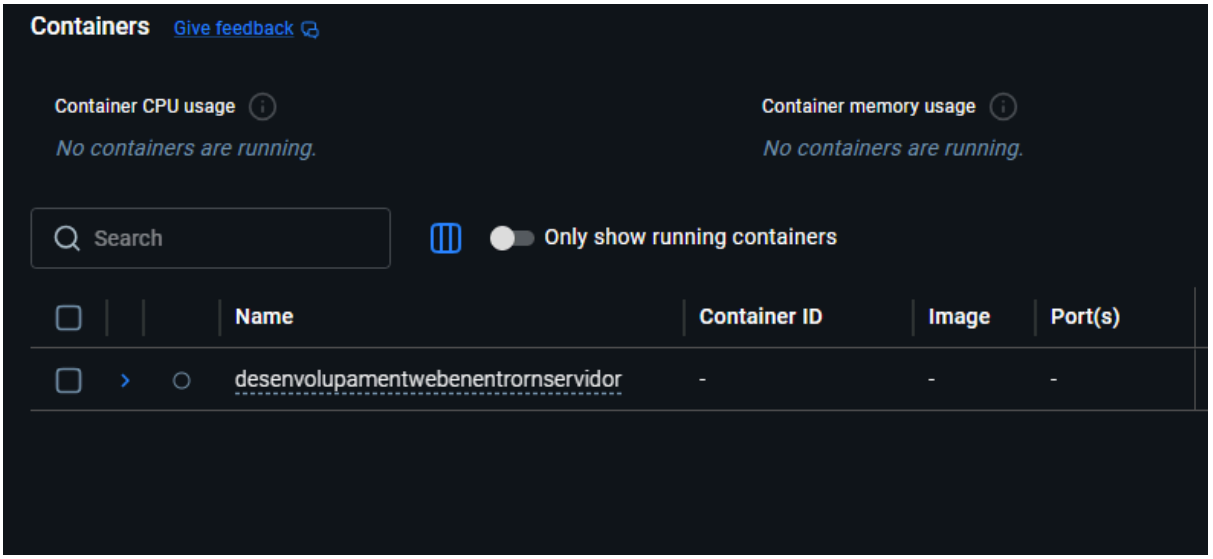
```
→ Practica 1 git:(development) X docker rm -f web-php mysql-db
web-php
mysql-db
→ Practica 1 git:(development) X docker rmi web-php web-php:backup mysql-db:backup
Untagged: web-php:latest
Deleted: sha256:f9cf95302e368403c1e686cd50f062bbf61b89a5157150d30fc39ab7b39d9262
Untagged: web-php:backup
Deleted: sha256:a6a903d8cfd76df5ff6067fa9dbf91cd8f603245b6151a684c5148766048b330
Deleted: sha256:f9cf95302e368403c1e686cd50f062bbf61b89a5157150d30fc39ab7b39d9262
Untagged: mysql-db:backup
Deleted: sha256:9a15dd0987e497d62a8de08cbb812fe1769fe029c3ccded5601b754bc022d5f9
→ Practica 1 git:(development) X sudo rm -rf mysql-data logs
→ Practica 1 git:(development) X docker images
```

```
→ Practica 1 git:(development) X docker images

IMAGE               ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest  c567b231f656       25.9kB         9.49kB
holahtml:latest     0737ac44bb95        221MB         53.8MB
mysql:8.0           7dcddc01f13b        1.1GB          249MB    U
php:8.2-apache      163407b1ceca        715MB          183MB    U
→ Practica 1 git:(development) X docker container ls -a
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS                   PORTS     NAMES
d09c4ff1565b   php:8.2-apache   "docker-php-entrypoi…"   6 weeks ago   Exited (0) 5 weeks ago             Dev_web_servidor
c960719a13e0   mysql:8.0        "docker-entrypoint.s…"   6 weeks ago   Exited (0) 5 weeks ago             desenvolupamentwebenentrornservidor-db-1
→ Practica 1 git:(development) X
```

```
→ Practica 1 git:(development) X docker images

IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest    c567b231f656       25.9kB         9.49kB
holahtml:latest       0737ac44bb95        221MB          53.8MB
mysql:8.0             7dcddc01f13b        1.1GB           249MB    U
php:8.2-apache        163407b1ceca        715MB           183MB    U
```



Per restaurarles:

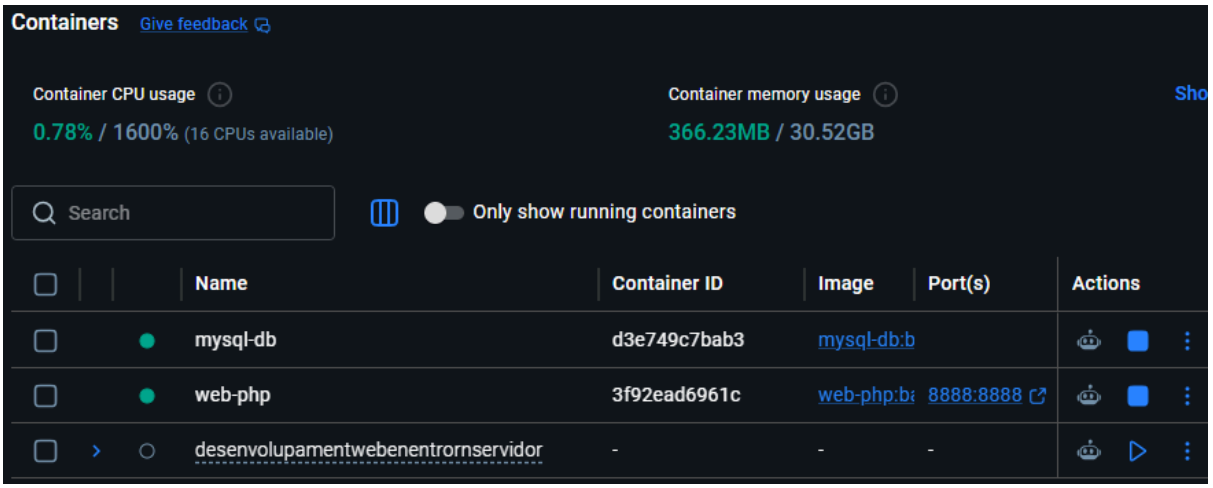
```
→ Practica 1 git:(development) X docker load -i backup/web-php-backup.tar
Loaded image: web-php:backup
→ Practica 1 git:(development) X docker load -i backup/mysql-db-backup.tar
Loaded image: mysql-db:backup
→ Practica 1 git:(development) X sudo tar xzf backup/mysql-data.tar.gz
→ Practica 1 git:(development) X sudo tar xzf backup/logs.tar.gz
→ Practica 1 git:(development) X docker run -d --name mysql-db -e MYSQL_ROOT_PASSWORD=pepito \
  -v "$PWD/mysql-data":/var/lib/mysql mysql-db:backup
d3e749c7bab38278a61e67084930d2a091d7cdfb20563a14482db1bcb9ffd5bc
→ Practica 1 git:(development) X docker run -d --name web-php -p 8888:8888 \
  -v "$PWD/logs":/var/log/apache2 web-php:backup
3f92ead6961c5ebb230784cc04d1e7e90a3b4fe17e9b830996c7bfceb486325d
→ Practica 1 git:(development) X
```

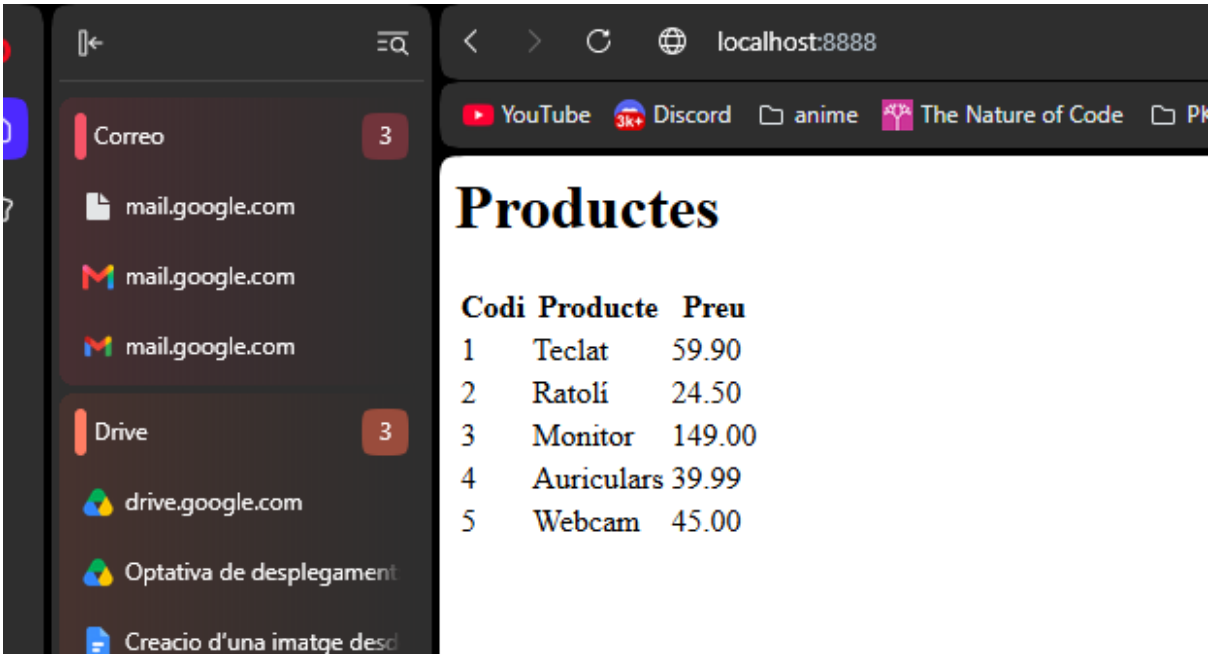
```
→ Practica 1 git:(development) X docker images

IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest    c567b231f656       25.9kB         9.49kB
holahtml:latest       0737ac44bb95        221MB          53.8MB
mysql-db:backup       9a15dd0987e4       1.09GB           234MB    U
mysql:8.0             7dcddc01f13b        1.1GB           249MB    U
php:8.2-apache        163407b1ceca        715MB           183MB    U
web-php:backup        a6a903d8cfd7        706MB           175MB    U
→ Practica 1 git:(development) X docker container ls -a
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS                      PORTS                                       NAMES
3f92ead6961c   web-php:backup     "docker-php-entrypoi…"   53 seconds ago   Up 53 seconds               0.0.0.0:8888->8888/tcp, [::]:8888->8888/tcp   web-php
d3e749c7bab3   mysql-db:backup    "docker-entrypoint.s…"   58 seconds ago   Up 58 seconds               3306/tcp, 33060/tcp                         mysql-db
d09c4ff1565b   php:8.2-apache     "docker-php-entrypoi…"   6 weeks ago      Exited (0) 5 weeks ago                                                  Dev_web_servidor
c960719a13e0   mysql:8.0          "docker-entrypoint.s…"   6 weeks ago      Exited (0) 5 weeks ago                                                  desenvolupamentwebenentrornservidor-db-1
→ Practica 1 git:(development) X
```

```
→ Practica 1 git:(development) X docker images

IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   c567b231f656       25.9kB         9.49kB
holahtml:latest      0737ac44bb95        221MB          53.8MB
mysql-db:backup      9a15dd0987e4       1.09GB          234MB   U
mysql:8.0            7dcddc01f13b        1.1GB          249MB   U
php:8.2-apache       163407b1ceca        715MB          183MB   U
web-php:backup       a6a903d8cfd7        706MB          175MB   U
```





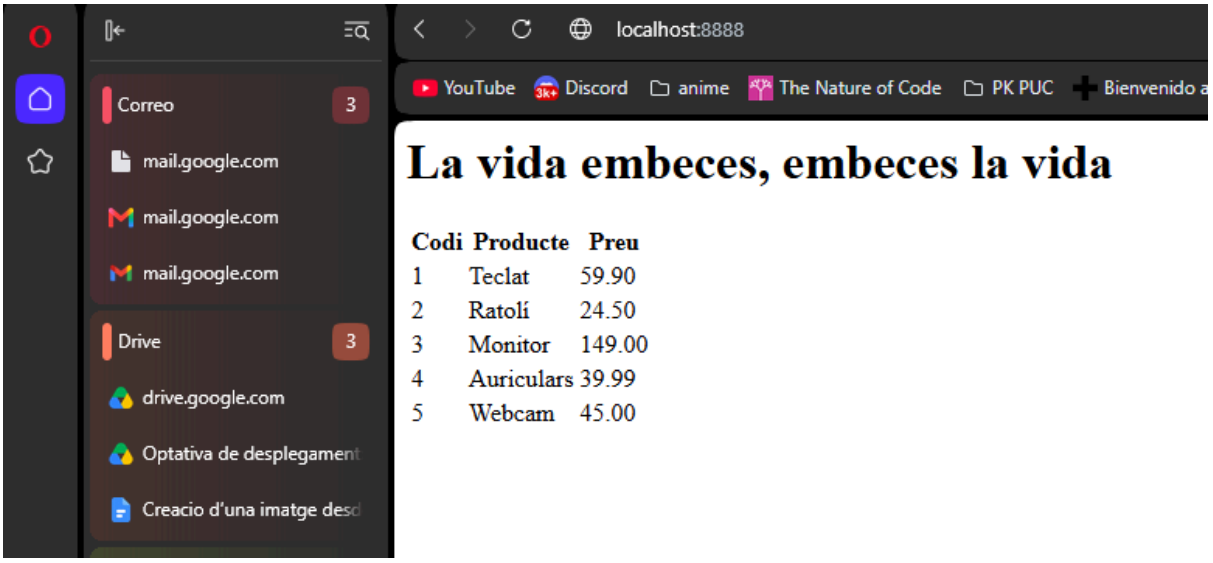
6. Ara que tot torna a funcionar, modifiquem algun fitxer de l'aplicació i pugem-lo la la màquina al contenidor que està funcionant.

A inndex.php li canviem:

sed -i 's|<h1>Productes</h1>|<h1>La vida embeces, embeces la vida</h1>|' index.php

```
→ Practica 1 git:(development) X sed -i 's|<h1>Productes</h1>|<h1>La vida embeces, embeces la vida</h1>|' index.php
→ Practica 1 git:(development) X
```

```
→ Practica 1 git:(development) X sed -i 's|<h1>Productes</h1>|<h1>La vida embeces, embeces la vida</h1>|' index.php
→ Practica 1 git:(development) X docker cp index.php web-php:/var/www/html/index.php
Successfully copied 1.35kB (transferred 3.07kB) to web-php:/var/www/html/index.php
→ Practica 1 git:(development) X
```



Dues comandes d'arrencada del servidor (en ordre)

1. Contenidor de la base de dades:

```
docker run -d --name mysql-db \
  -e MYSQL_ROOT_PASSWORD=pepito \
  -v "$PWD/mysql-data":/var/lib/mysql \
  -v "$PWD/bd.sql":/docker-entrypoint-initdb.d/bd.sql:ro \
  mysql:8.0
```

2. Contenidor de la web:

```
docker run -d --name web-php \
  -p 8888:8888 \
  -v "$PWD/logs":/var/log/apache2 \
  web-php
```